AFFINENESS APPEARS AT A FINITE STAGE UNDER AFFINE TRANSITIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1583-1589, following the statement at 1570-1581.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Affine is etale-local on the target. Morphisms of Spaces, Lemma spaces-morphisms-lemma-affine-local (cross-chapter, cited at ~1585-1587). It suffices to check X_i x_Y W -> W affine after a surjective etale W -> Y.
- Affineness of the limit appears at a finite stage. lemma-limit-is-affine (same chapter, card L26293, cited at ~1588-1589). Concluding engine on the charted system.

Source reference. Original lemma 07SQ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1370–1370. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03WG: focused-seeds/geometry/S9-spaces-limits-eventually-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let Y be an algebraic space over S. Let $X = \lim X_i$ be a directed limit of algebraic spaces over Y with affine transition morphisms. Assume*

(1) *Y quasi-compact and quasi-separated,*
(2) *X_i quasi-compact and quasi-separated,*
(3) *$X \to Y$ affine.*

Then $X_i \to Y$ is affine for i large enough.

Proof. Choose an affine scheme W and a surjective étale morphism $W \to Y$. Then $X \times_Y W$ is affine and it suffices to check that $X_i \times_Y W$ is affine for some i (Morphisms of Spaces, Lemma 03WG). This follows from Lemma 07SQ. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.